

High accuracy without utility? The paradox of Bitcoin AI forecasts

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Supplemental Data and Research Materials

Supplemental data is provided by the corresponding author and can be obtained on request.

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Abstract

Although previous artificial intelligence models achieve high statistical accuracy and low error metrics when predicting absolute Bitcoin prices, our findings reveal a critical paradox: their forecasting utility remains limited. This limitation is largely due to strong autocorrelation effects and a tendency to extrapolate recent trends rather than anticipate meaningful market changes.

We show that, in contrast, predictions of daily returns, despite weaker statistical performance, offer greater potential for forecasting. To capture the complexity of the cryptocurrency market, we incorporate sentiment indicators, technical analysis metrics, exchange-level data, and macroeconomic variables. We also introduce an ensemble approach that combines Long Short-Term Memory (LSTM) networks with Random Forest predictions. This method improves directional accuracy, reaching 62.67 percent during the test period and outperforming both models individually.

The findings underline the value of ensemble models in cryptocurrency forecasting and underscore the importance of evaluating predictive models based on performance rather than statistical fit alone.

Keywords: LSTM, Bitcoin, AI, Forecasting, Cryptocurrencies

1. INTRODUCTION

Since its introduction by Satoshi Nakamoto in 2008, Bitcoin (BTC) has evolved from a decentralized digital currency concept into a major speculative financial asset, drawing increasing interest from investors, regulators, and the general public. Key milestones such as the late 2020 bull run and the approval of the first spot Bitcoin ETF in 2024 have further cemented its relevance and growing acceptance in mainstream financial markets (Mazur & Polyzos, 2024). Unlike traditional financial assets, Bitcoin is not backed by physical commodities or centralized institutions. Its value stems from cryptographic security and decentralized trust mechanisms (Härdle et al., 2020; Shahzad et al., 2018). This fundamental distinction introduces substantial challenges for price forecasting, as conventional valuation frameworks often fall short in capturing the unique dynamics of cryptocurrency markets (Corbet et al., 2019).

In light of Bitcoin's high volatility and increasing financial integration, the demand for accurate and reliable forecasting models has intensified. Among advanced techniques, deep learning approaches, particularly Long Short-Term Memory (LSTM) networks, have shown considerable promise in time series forecasting and have gained traction in financial research (Karim et al.,

2018). In the context of cryptocurrencies, LSTM models have frequently outperformed traditional machine learning methods. Recent studies report encouraging results when using LSTM models to predict price movements of Bitcoin and other major cryptocurrencies. For instance, Yiying and Yeze (2019), Seabe et al. (2023), and Birim (2022) report low error metrics for LSTM-based forecasts using historical price data. Hansun et al. (2022) and Birim (2022) also find that LSTM and GRU models outperform simpler approaches based on indicators such as RMSE and MAPE. Similarly, Liu et al. (2021) highlight the strong performance of deep learning models such as Stacked Denoising Autoencoders (SDAEs) for Bitcoin price forecasting. However, these studies largely assess model success through statistical performance indicators, often without questioning their real-world applicability or interpretability.

This study builds on a critical paradox in AI-driven cryptocurrency forecasting: although LSTM models achieve excellent error metrics when predicting Bitcoin closing prices, this success is largely driven by strong autocorrelation in the data rather than by meaningful foresight. Absolute price forecasts often reflect the mere continuation of existing trends, offering limited practical value for investors and decision-makers. In contrast, predictions of daily returns, despite involving higher numerical errors, tend to capture more relevant market dynamics and offer greater potential for actionable insights.

To address this, we develop forecasting models that incorporate a diverse set of features, including sentiment indicators, technical analysis metrics, macroeconomic variables, and crypto exchange-level data. This integration allows for a more comprehensive reflection of the cryptocurrency market's complexity and volatility. Furthermore, we propose and evaluate an ensemble architecture that combines LSTM networks with Random Forest predictions. This hybrid approach significantly improves directional accuracy, achieving approximately 63 percent correctness in predicting daily return movements, and outperforms the individual models. These findings underscore the potential of combining sequential and structural learning for forecasting fast-moving and complex assets like Bitcoin.

Future research should aim to expand the methodology to a wider range of digital assets, incorporate richer real-time data sources such as social media sentiment, and refine ensemble models through attention mechanisms or transformer-based architectures. Additionally, greater emphasis should be placed on evaluation metrics that reflect forecasting utility rather than relying exclusively on statistical error measures. By doing so, we expect AI-based forecasting models can evolve beyond trend replication and toward offering genuine value in financial decision-making.

2. DATA COLLECTION AND PREPROCESSING

Our study relies on a comprehensive dataset that integrates technical indicators, sentiment measures, macroeconomic variables, and crypto exchange-specific data to capture the complexity of Bitcoin price movements. Bitcoin price and volume data were primarily retrieved from Yahoo Finance, while macroeconomic indicators such as the S&P500 Index, Gold prices, Volatility Index (VIX), USD Index, and the 5-Year Forward Inflation Expectation Rate were collected via Yahoo Finance and the Federal Reserve Economic Data (FRED) API.

Investor sentiment was incorporated through multiple sources, including the Crypto Fear & Greed Index, Google Trends data for selected Bitcoin-related search terms, and the News Sentiment Index published by the San Francisco Fed. To capture crypto-specific market dynamics, trading data from Binance, including funding rates, open interest, long/short ratios, and liquidation volumes, was gathered using the Coinalyze API.

Technical indicators were computed using the 'pandas-ta' library. Key features included moving averages (SMA50, SMA200, EMA50), momentum and trend indicators (MACD, RSI, Stochastic Oscillator, ROC, CCI, ADX), and volatility measures (Bollinger Bands, ATR). Sentiment features captured market fear, greed, and search interest, while macroeconomic indicators represented broader economic conditions affecting Bitcoin.

Given the wide range of features, a feature selection process was necessary to enhance model efficiency and prevent overfitting. We employed SHapley Additive exPlanations (SHAP) derived from an Extra Trees Regressor to rank feature importance (cf. Goodell et al., 2023). Based on this analysis, the top ten most relevant features were selected separately for price and return prediction models. Features differed notably between tasks, with technical and sentiment indicators dominating return prediction, while absolute price models relied more heavily on historical price data and macroeconomic signals.

The dataset was limited to the period from October 2020 to November 2024 to ensure the availability of all variables, particularly crypto exchange indicators. Missing values, especially in weekly Google Trends data, were forward-filled, and all features were normalized using Min-Max scaling to improve model training stability.

For model development, we employed two architectures: (1) a two-layer stacked LSTM network with dropout regularization for overfitting prevention, and (2) a variant with an attention mechanism based on Abbasimehr and Paki (2020) to enhance long-term dependency capture. As the second approach produced better results, we chose this architecture for our final LSTM model.

Hyperparameter optimization was conducted via grid search, adjusting neuron counts, dropout rates, batch sizes, and optimizers to achieve the best performance.

The dataset was split chronologically into training (95%) and test (5%) sets, preserving temporal structure critical for time series forecasting. Evaluation metrics included Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Directional Accuracy (MDA), R^2 goodness-of-fit, and Pearson correlation. Special emphasis was placed on MDA to assess the models' ability to predict the correct direction of price movements – a key measure of practical utility beyond mere numerical accuracy.

3. RESULTS

We evaluated the forecasting performance of LSTM models for Bitcoin closing prices and daily returns and compared them against alternative machine learning models, baseline methods, and ensemble approaches.

3.1. Closing Price Prediction

The LSTM model achieved strong performance in forecasting Bitcoin closing prices. It recorded a Mean Absolute Error (MAE) of \$1865, a Root Mean Squared Error (RMSE) of \$3291, a Mean Absolute Percentage Error (MAPE) of 2.61%, and a correlation coefficient of 0.94 indicating a strong interconnection between predictions and true values.

Visual inspection further supported this impression: the predicted price line closely tracked the actual Bitcoin prices across the test period. However, a closer look revealed a crucial limitation. Predictions often lagged behind real prices by one day (Appendix A, Figure 1). Supporting this interpretation, a simple baseline model, using today's closing price as tomorrow's forecast, achieved even lower error metrics.

Autocorrelation analysis confirmed this behavior. The first-lag autocorrelation (ACF(1)) for Bitcoin closing prices was exceptionally high at ~ 0.998 , meaning that yesterday's price almost perfectly predicts today's price. In such a setting, minimizing error metrics can be achieved simply by extrapolating the last observed value. Thus, while LSTM models achieve impressive statistical results in closing price forecasting, their practical forecasting utility appears highly limited.

3.2. Daily Return Prediction

In contrast, forecasting daily returns proved substantially more challenging. The LSTM model achieved a MAE of 1.87% (return units), a RMSE of 2.61%, and a MAPE of 186.74%, while directional prediction (MDA) was modest at 52%. Correlation between predicted and actual returns was close to zero at 0.15.

Unlike closing prices, Bitcoin daily returns exhibited no meaningful autocorrelation ($ACF(1) \sim 0.056$), making naive trend-extrapolating models ineffective. Consequently, even sophisticated LSTM models struggled to achieve strong predictive performance. High volatility and sudden movements remained challenging to capture, as seen in the return prediction plots (Appendix B, Figure 2). However, compared to the baseline, the LSTM model offered significant improvements in MAE and MDA, indicating that it captured some underlying structure beyond pure randomness.

3.3. Performance Comparison with Other Machine Learning Models

To assess the relative performance of LSTM models, we compared them against Random Forest (RF) and XGBoost regressors trained on the same feature sets. The detailed comparison of model performance across different architectures is summarized in Appendix, Table 1 for price forecasts and Table 2 for return forecasts.

For closing price forecasting, LSTM models clearly outperformed RF and XGBoost across all metrics. This is expected, given the strong sequential dependencies in price data that favor recurrent models like LSTM.

In contrast, for daily return forecasting, Random Forest models achieved a higher MDA (57%) than LSTM (52%), with slightly better absolute error metrics (MAE and RMSE). XGBoost performed similarly to RF. These results indicate that non-sequential models might be better suited for highly volatile and less autocorrelated targets like cryptocurrency returns.

Furthermore, a classical ARIMA model, often used for financial time series, performed poorly compared to both LSTM and tree-based models, confirming the inadequacy of traditional linear models in handling the non-linear dynamics of cryptocurrency returns.

3.4. Ensemble Model: Combining LSTM and Random Forest

Given these complementary strengths, we developed an ensemble model that combined LSTM and Random Forest predictions. Random Forest forecasts were included as additional input features into the LSTM model, aiming to leverage both structure-based learning and sequential dependency modeling. The exact results of these models for price and return forecasts are shown graphically in Appendix C.

For closing price prediction, ensemble modeling did not substantially improve performance compared to standalone LSTM models. However, for daily return prediction, the ensemble model achieved a significant increase in directional prediction accuracy. The MDA improved from 52% (LSTM) and 57% (Random Forest) to ~63% in the ensemble model, demonstrating a meaningful enhancement in practical forecasting utility.

These results highlight that combining different learning paradigms, in our case sequential pattern recognition via LSTM and structural pattern extraction via Random Forest, can create meaningful synergies, particularly in challenging low-predictability environments such as cryptocurrency returns.

4. CONCLUSION

This study highlights a critical paradox in AI-driven cryptocurrency forecasting: while LSTM models achieve excellent error metrics when predicting Bitcoin closing prices, this success is largely a result of the strong autocorrelation in price data rather than genuine foresight. Absolute price forecasts often merely extrapolate existing trends, offering limited practical value for market participants. In contrast, predicting daily returns, despite involving higher numerical errors, captures more meaningful dynamics and shows greater potential for actionable decision support.

Through the integration of sentiment indicators, technical analysis features, macroeconomic data, and crypto exchange metrics, we developed models that better reflected the underlying volatility and complexity of the crypto market. Particularly, the ensemble approach combining LSTM networks and Random Forest models significantly enhanced directional accuracy, achieving ~63% correctness in predicting daily return movements. This demonstrates that hybrid architectures, which blend sequential and structural learning, offer a promising path forward for forecasting volatile and rapidly evolving assets like Bitcoin.

Nonetheless, the study also faces limitations. Focusing exclusively on Bitcoin restricts the generalizability of our findings, and limited access to high-frequency and social media sentiment data constrained the breadth of inputs. Moreover, computational constraints limited the exploration of more complex architectures and larger-scale optimizations.

Future research should address these limitations by expanding analyses to multiple cryptocurrencies, integrating richer real-time sentiment signals, and refining ensemble and attention-based deep learning models. Additionally, greater emphasis should be placed on developing evaluation metrics that better capture real-world utility rather than relying solely on traditional error measures. By doing so, we can move closer to building AI models that truly add value in financial forecasting rather than simply reproducing historical trends.

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APPENDIX

Appendix A: Model Performances on Closing Price Prediction

	MAE (in %)	RMSE (in \$)	MAPE (in %)	MDA (in %)	Correlation
LSTM	1865.13	3291.04	2.6055	56.00	0.9401***
Baseline Model	1218.13	1795.07	1.8192	57.3333	0.9729***
RF Regressor	2251.05	4142.41	3.1787	50.6667	0.8459***
RF Classifier	-	-	-	53.3333	-
XGB Regressor	2966.17	4790.44	4.2443	49.3333	0.8226***
Ensemble Model	2388.07	3988.17	2.8142	57.3333	0.9149***

*, **, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 1: Error metrics of the closing price models

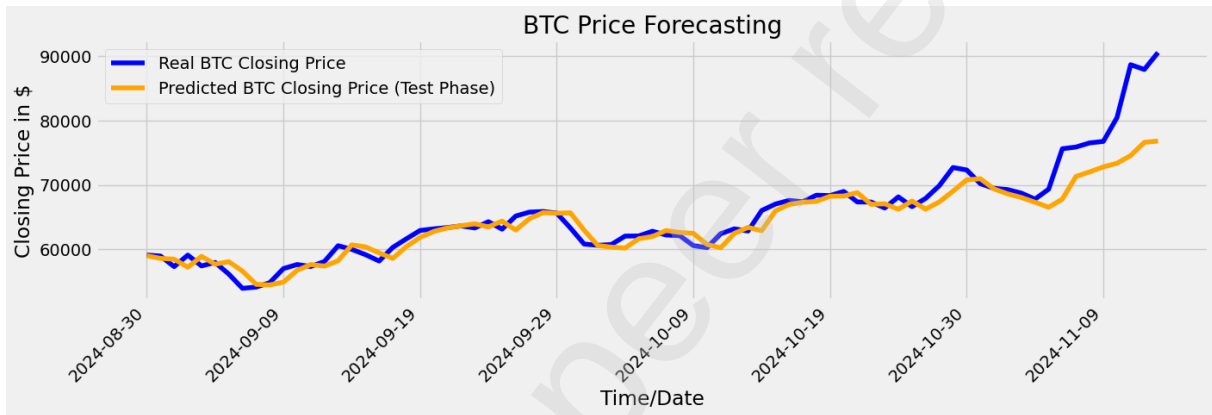


Figure 1: Visualization of prediction results and real BTC prices during out of sample testing phase

Appendix B: Model Performances on Daily Return Prediction

	MAE (in % ¹)	RMSE (in % ¹)	MAPE (in %)	MDA (in %)	Correlation
LSTM	1.8697	2.6068	186.74	52.00	0.1537
Baseline Model	2.5666	3.3048	669.6773	36.00	0.1459
RF Regressor	1.9549	2.5331	391.2097	57.3333	0.1840
RF Classifier	-	-	-	56.00	-
XGB Regressor	2.2621	2.9039	736.7786	57.3333	0.0885
ARIMA	2.0472	2.8369	161.6514	2.0134	0.0036
Ensemble Model	1.9918	2.5458	409.2333	62.67	0.2345**

¹ % is here the unit of the target variable, and is not the same as the relative information in MAPE or in MDA

*, **, *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 2: Error metrics of the daily return models

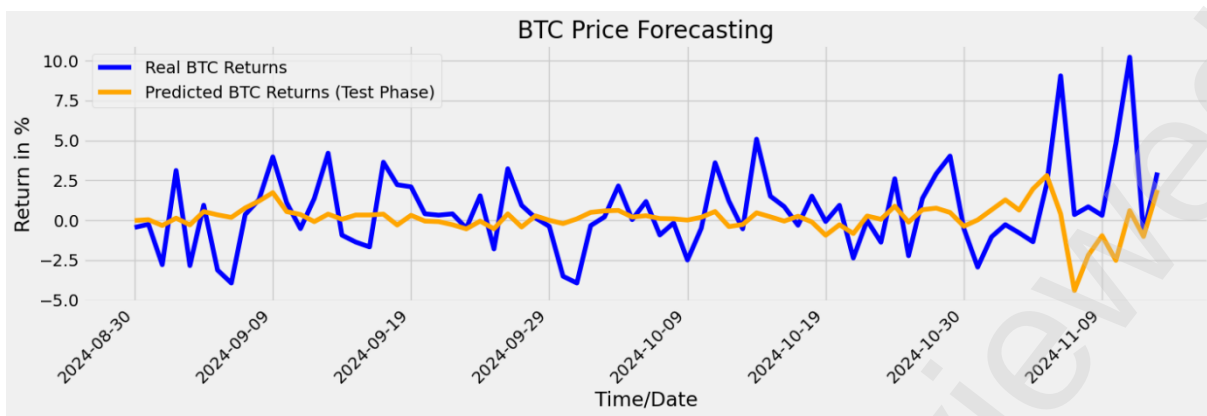


Figure 2: Visualization of LSTM prediction results and real BTC daily returns during testing phase

Appendix C: Ensemble Model Performance

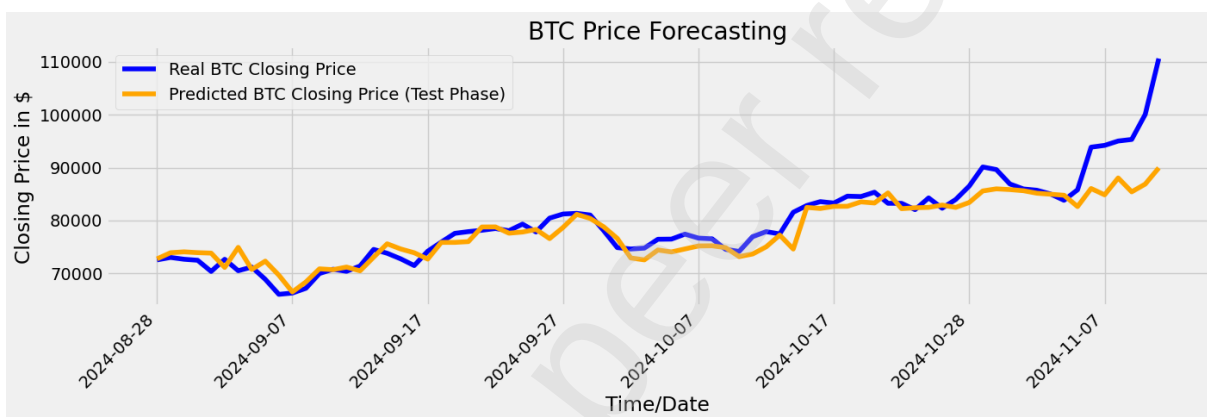


Figure 3: Price prediction with Ensemble Model

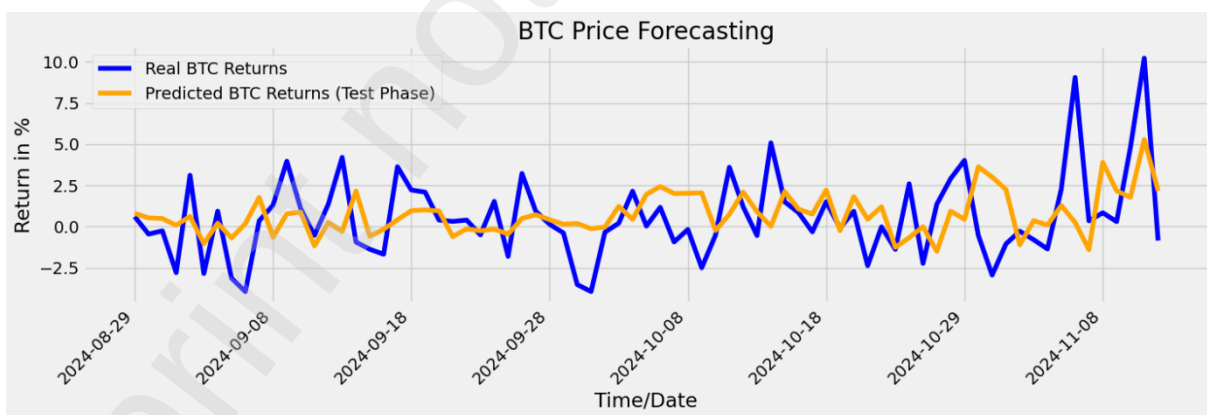


Figure 4: Daily Return prediction with Ensemble Model